PRACTICAL NO.3(A)

Aim:- Create stacked and grouped bar charts for comparision.

Dataset:

Abc Sheet1 Student ID	Abc Sheet1 Department	Abc Sheet1 Activity	# Sheet1 Hours	# Sheet1 Attendance	# Sheet1 Score	Abc Sheet1 Activity Date
S113	BSc IT	Sports	13	94	91	22-08-2024
S114	BSc CS	NSS	15	81	56	28-08-2024
S115	Bcom	Music	17	97	78	5/9/2024
S116	BSc IT	Drama	8	88	86	12/9/2024
S117	BSc CS	Sports	10	83	48	18-09-2024
S118	Bcom	NSS	16	95	57	25-09-2024
S119	BSc IT	Music	12	90	68	2/10/2024
S120	BSc CS	Drama	6	84	95	10/10/2024
S121	Bcom	Sports	9	91	84	18-10-2024
S122	BSc IT	NSS	10	77	76	22-10-2024
S123	BSc CS	Music	15	48	53	20-11-2024
S124	Bcom	Drama	13	98	84	13-11-2024
S125	BSc IT	Sports	14	78	98	25-11-2024
S126	BSc CS	NSS	9	86	86	27-11-2024
S127	Bcom	Music	8	84	84	3/12/2024
S128	BSc IT	Drama	12	75	59	11/12/2024
S129	BSc CS	Sports	13	73	76	13-12-2024
S130	Bcom	NSS	12	87	86	20-12-2024
S131	BSc IT	Music	10	88	88	125-12-2024
S132	BSc CS	Drama	8	90	77	28-12-2024
S133	Bcom	Sports	8	95	84	1/1/2025
S134	BSc IT	NSS	9	96	75	12-01-205
S135	BSc CS	Music	10	80	66	23-01-2025

Objective:-
Compare total activity hours of each department, divided by activity type.

a.) What is stacked bar chart ?
A stacked bar chart is a type of bar chart in which each bar is divided into different colored segments. Each segment represents a category, and together they show both the total value and contribution of each category.

Steps:-

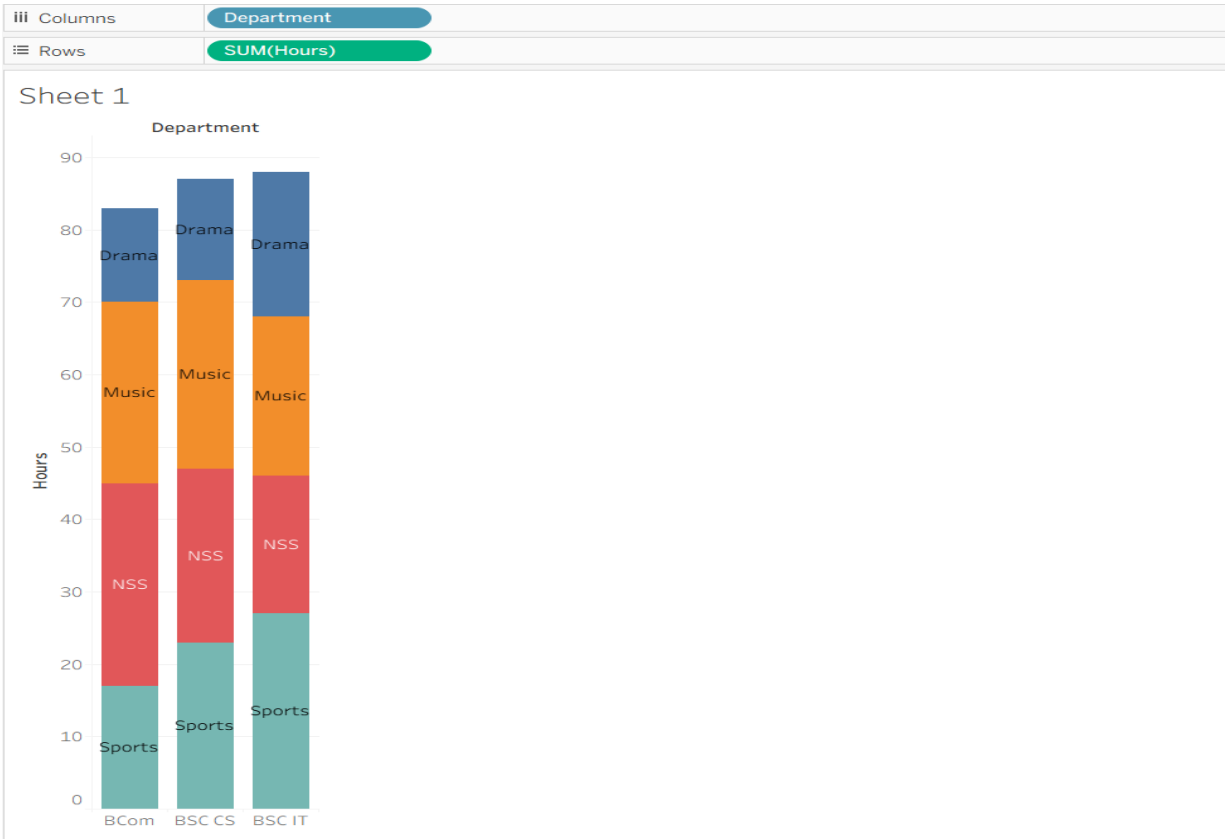
Step 1:
Drag **Department → Column**

Step 2:
Drag **Hours → Rows**
A bar chart appears

Step 3:
Drag **Activity → Color(Marks Card)**
Each department bar is divided into different colors.

Step 4:
If bars are not stacked
Click:
Analysis → Stack Marks → On

Output:-
Each department(BSC IT, BSC CS, BCom) shows total hours divided into sports, music, drama and NSS.



b.) What is grouped bar chart ?

A grouped bar chart is also called side-by-side bar chart, is a chart that displays multiple bar chart next to each other for each category. It is used to compare the values of different groups within the same category.

Objective:-
Compare Activity Hours department wise.

Steps:-

Step 1:
Open **new worksheet**

Step 2:
Drag **Department → Columns**

Step 3:
Drag **Activity → Column**
(Place after Department)

Step 4:
Drag **Hours → Rows**

Step 5:
Drag **Activity → Color**
Sepearte color appears for each activity.

Step 6:
Drag **Activity → Label**
Name of each activity appears

Output:-

Each department displays separate bars for Sports, Music, Drama & NSS.



PRACTICAL NO. 3(B)

Aim:- Build Treemaps and pie chart for part to whole analysis.

a.) What is a treemap ?

A treemap is a visualization that displays hierarchical or categorical data as nested rectangles. The size of each rectangle represents the value of measure (such as Hours or Score), while the color can represent another measure or category.

Objective:-
Show which activity takes more student hours.

Steps:-

Step 1:-
Open new worksheet

Step 2:-
Click → **Show Me**

Step 3:-
Click → **TreeMap**

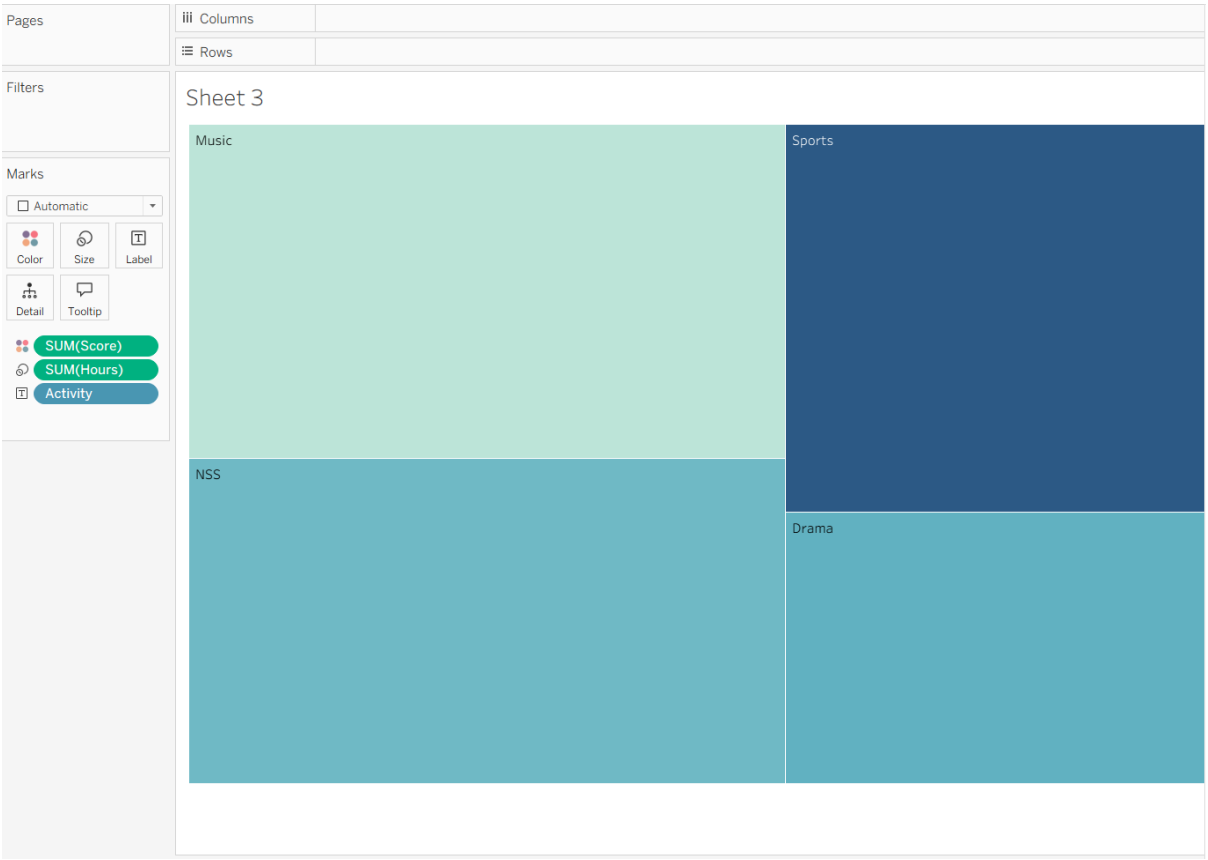
Step 4:-
Drag **Activity → Label**

Step 5:
Drag **Hours → Size**

Step 6:
Drag **Score → Color**

Output:-

Large rectangle represents activities with higher total hours.



b.) What is pie chart ?

A pie chart shows the relationship of parts to the whole for a categorical variable by depicting a circle, or pie, divided into segments. The size of each segment, or slice of the pie, represents the proportional contribution of a specific category to the whole.

Objective:-
Show percentage by activity.

Steps:-

Step 1:-
Open **new worksheet**

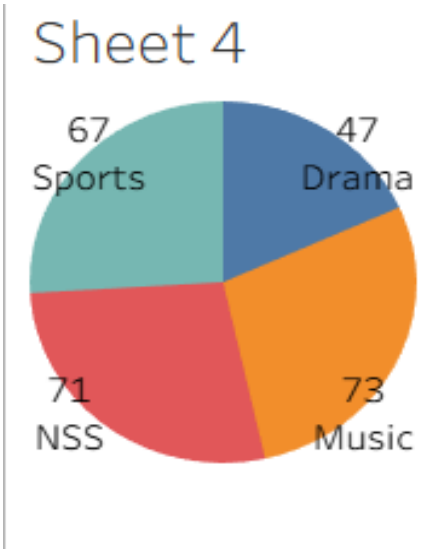
Step 2:
Choose → **PieChart**

Step 3:
Drag **Activity** → **Color**

Step 4:
Drag **Hours** → **Angle**

Step 5:
Drag **Hours** → **Label**

Output:-
Pie Chart showing each activity's share of total hours.



PRACTICAL NO. 3(C)

Aim:- Design Scatter plot to identify relationships.

a.) What is scatterplot ?

A Scatterplot is a graph that displays the relationship between two numerical variables using points. Each point represents one record in the dataset.

Objective:-
Find relationship between attendance and score.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Attendance → Column**

Step 3:-
Drag **Score → Rows**
An empty scatterplot appears on screen.

Step 4:-
Drag **Department → Color**
4 colors appears on screen each indicating specific department

Step 5:-
Drag **StudentID → Detail**
Multiple small circles appear on screen with specific color of their activity each representing a student

Output:-

Each student appears as a point

Student with higher attendance generally have higher score



PRACTICAL NO. 3(D)

Aim:- Create dual axis and combination chart.

b.) What is dual- axis chart ?

A Dual axis chart is a chart that displays two different measures on the same graph, each with its own axis. It allows you to compare two related measures over the same dimension(such as time).

Objective:-
Compare Hours and Score together.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Activity Date → Column**
(Choose Month)

Step 3:-
Drag **Hours → Rows**

Step 4:-
Drag **Score → Rows**
(Place besides hours)
Two graphs will appear

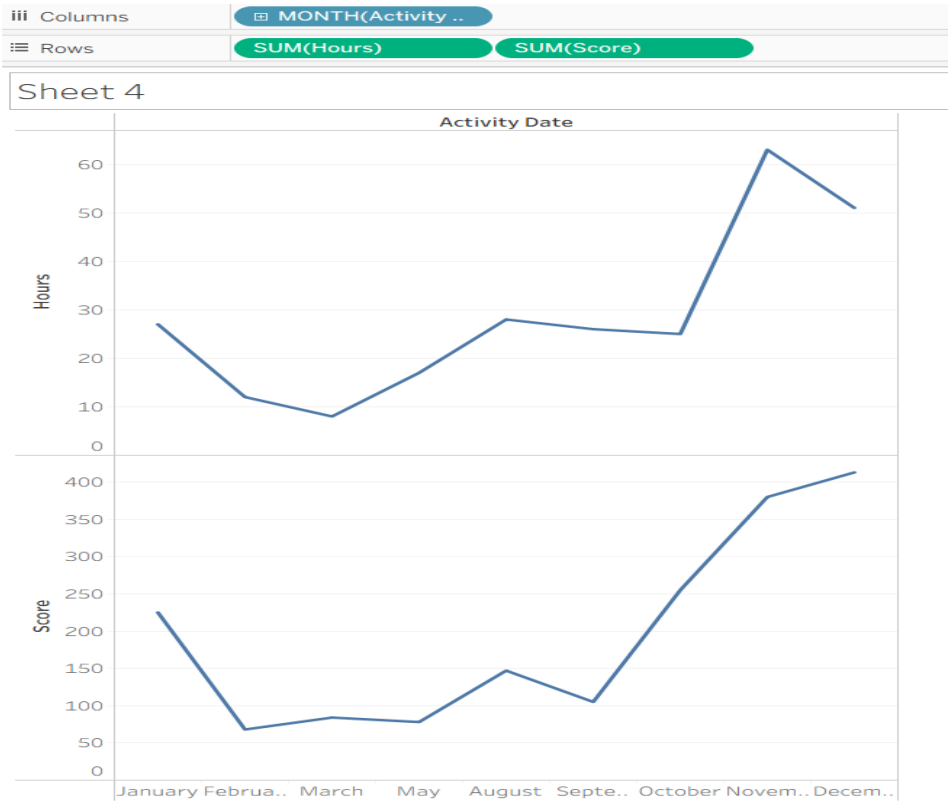
Step 5:-
Right Click
Score **Axis → Dual Axis**

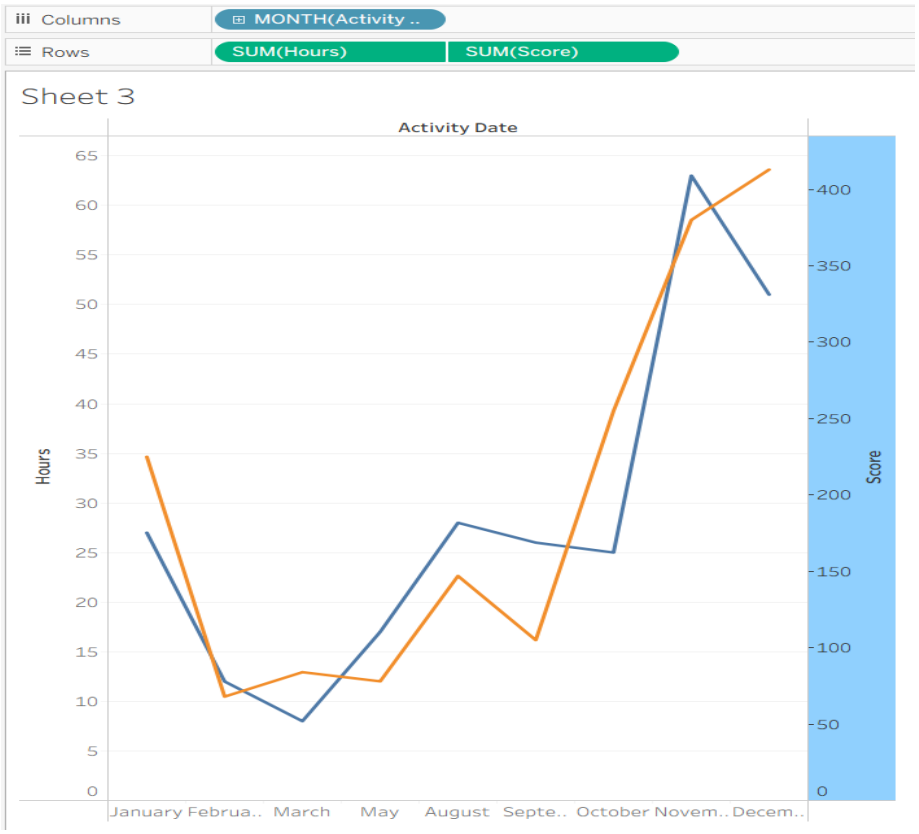
Step 6:-
Right **Axis → Synchronize Axis**

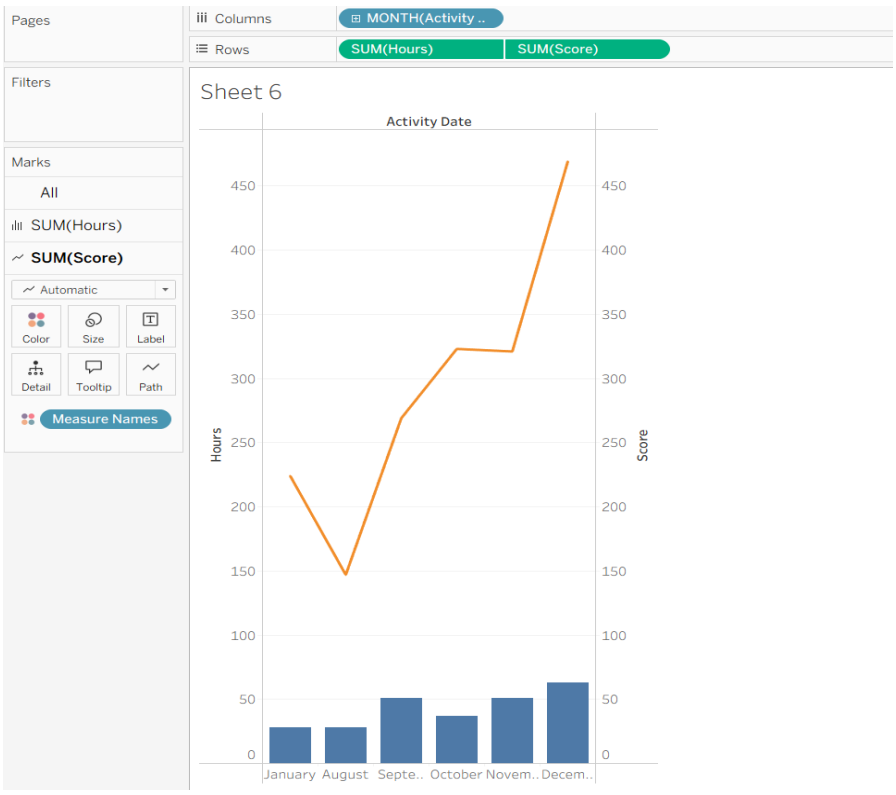
Step 7:-
Marks Card
Hours → Bar

Marks Card
Score → Line

Output:-
Bars shows Hours
Line shows Score




Columns
MONTH(Activity ..
Rows
SUM(Hours)
SUM(Score)
Sheet 3
Activity Date
Hours
Score
65
60
55
50
45
40
35
30
25
20
15
10
5
0
400
350
300
250
200
150
100
50
0
January
Februa..
March
May
August
Septe..
October
Novem..
Decem..


Pages
Columns
MONTH(Activity ..
Rows
SUM(Hours)
SUM(Score)
Filters
Marks
All
SUM(Hours)
SUM(Score)
Automatic
Color
Size
Label
Detail
Tooltip
Path
Measure Names
Sheet 6
Activity Date
Hours
Score
450
400
350
300
250
200
150
100
50
0
January
August
Septe..
October
Novem..
Decem..

PRACTICAL NO. 3(E)

Aim:- Develop histogram for distribution analysis.

a.) What is histogram ?

A histogram is a chart that shows the distribution(frequency) of a numerical variable by grouping values into continuous ranges called bins.

Objective:-
See the distribution of student scores.

Steps:-

Step 1:-
Open new worksheet

Step 2:-
Right Click
Score → Create → Bins

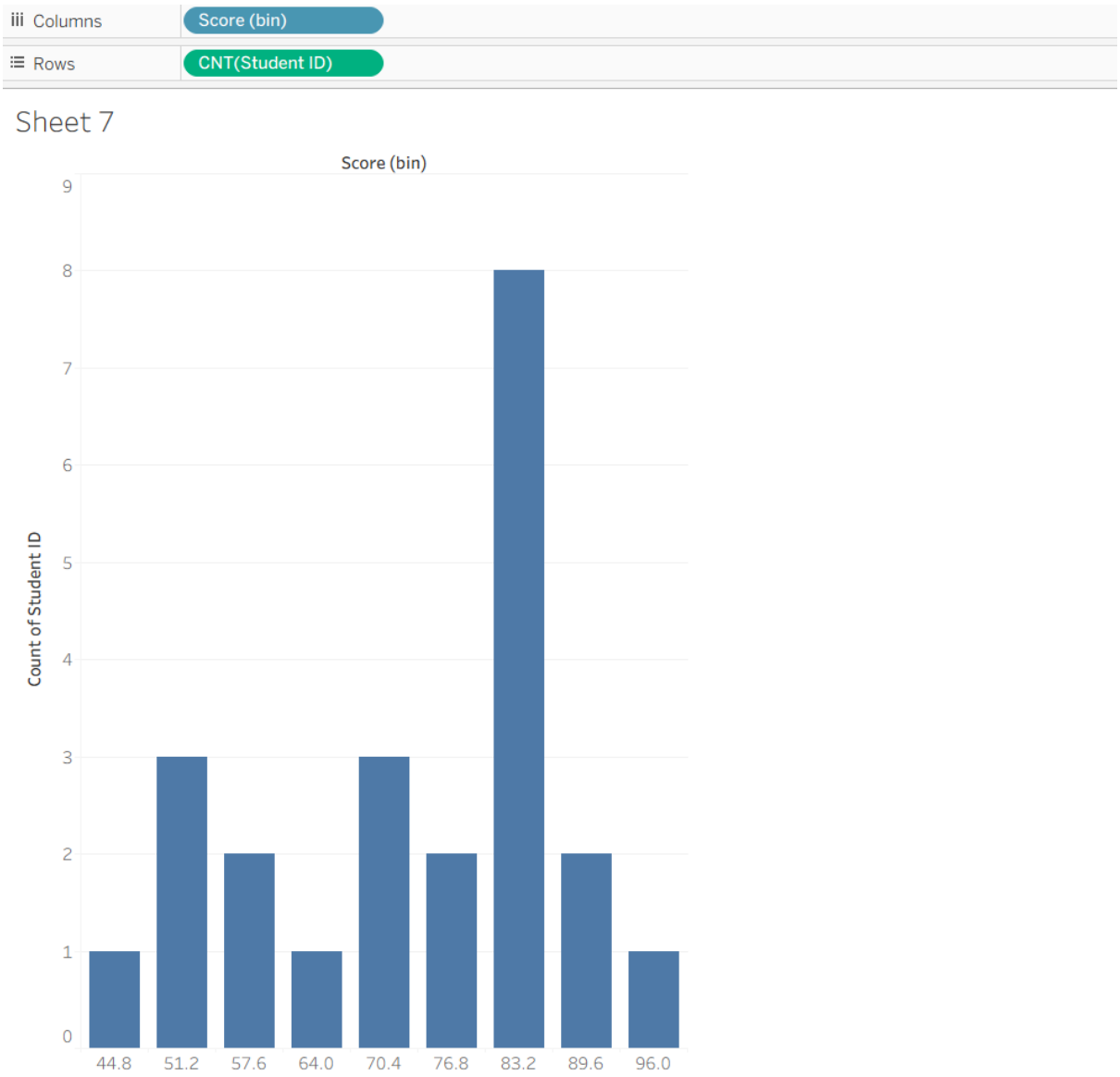
Step 3:-
Accept **default size → Ok**
A new field
Score(bins)
Appears

Step 4:-
Drag **Score(bins) → Column**
Different scores on the screen

Step 5:-
Drag **Number of record → Rows**
Histogram will appear on screen

Output:-

Histogram showing how many students fall in each score range



PRACTICAL NO. 3(F)

Aim:- Build box and whisker plots.

a.) What is box and whisper plot ?

A box and whisper plot, also called a Box Plot, is a chart that summarizes the distribution of numerical data using the minimum value, first quartile(Q1), median, third quartile(Q3), and maximum value. It can also highlight outlier.

Objective:-
Compare score distribution across department

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Department → Column**
The Department field is added to the Columns shelf, creating separate categories.

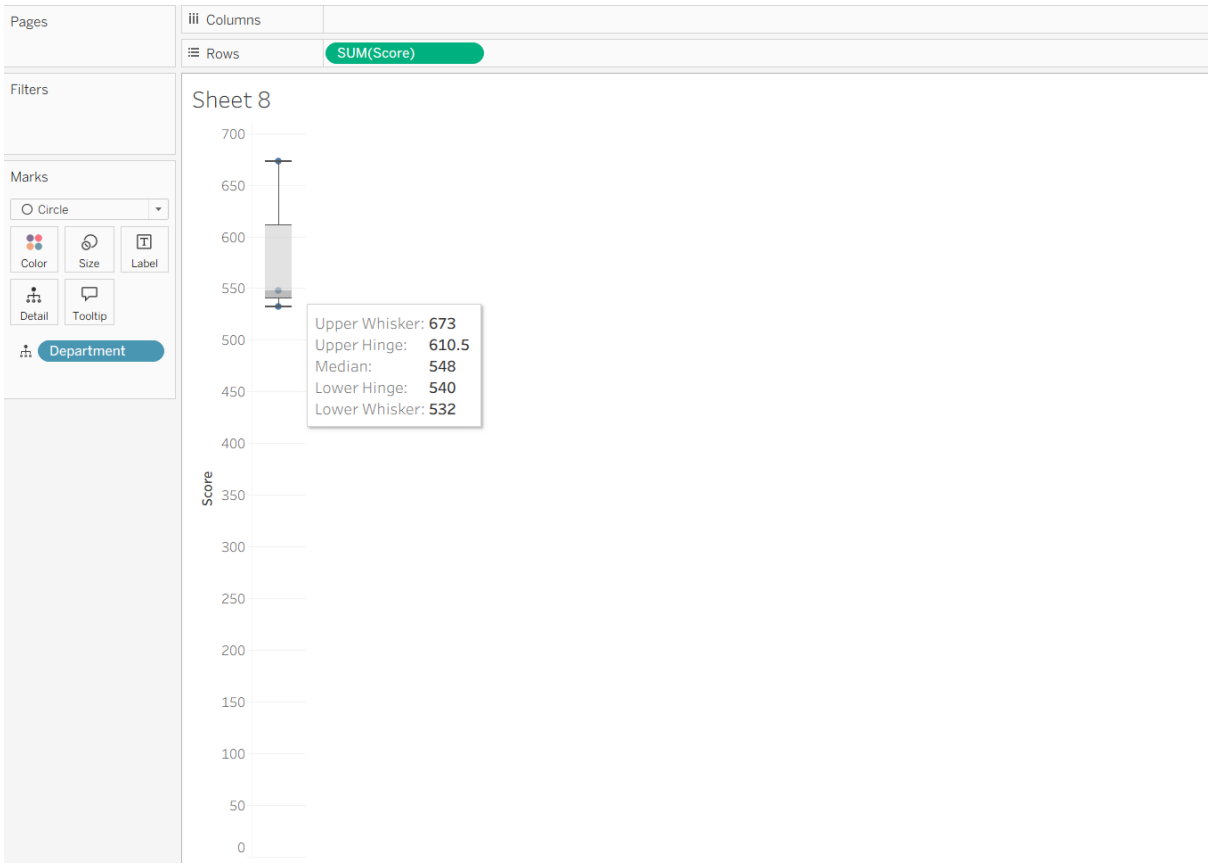
Step 3:-
Drag **Score → Rows**
The Score field is added to the Rows shelf, displaying score values for each department.

Step 4:-
Click **Show me → Box and whisper plot**
Tableau generates a Box-and-Whisker Plot showing the distribution, median, spread, and outliers of scores for each department.

Output:-

One box for each department showing:

- Minimum Score
- Maximum Score
- Median
- Quartiles
- Outliers(If any)



PRACTICAL NO. 3(G)

Aim:- Use gantt chart and advance date visualization.

a.) What is gantt chart ?

A gantt chart is a chart that displays tasks or activities over time using horizontal bars. The position of each bar shows when the activity starts, and the length of bar represents its duration.

Objective:-
Display activity duration using Hours.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **StudentID → Rows**

Step 3:-
Drag **Activity Date → Column**
Choose Exact Date

Step 4:-
Marks → Gantt Chart

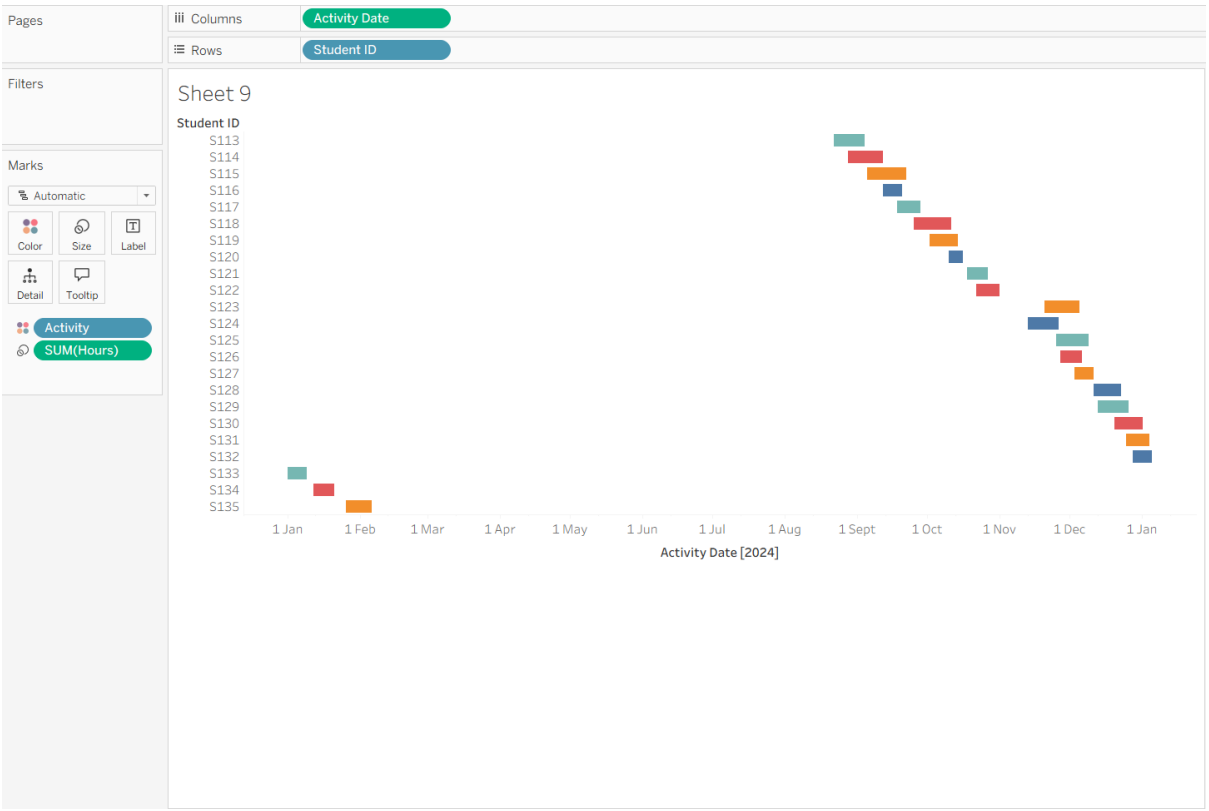
Step 5:-
Drag **Hours → Size**

Step 6:-
Drag **Activity → Colors**

Output:-

Each student has a Gantt bar.

Longer bars represents more activity hours.



Conclusion:

This practical demonstrated how different Tableau charts can be used to visualize and analyze data effectively for various purposes. It helped in comparing values, identifying relationships, understanding distributions, and presenting data in a clear and meaningful way using appropriate chart types. Overall, these visualizations improve data interpretation and support better decision-making.